

Derivative Example

Cost function $J(w) = w^2$

Say $w = 3$ $J(w) = 3^2 = 9$

If we increase w by a tiny amount $\epsilon = 0.001$ how does $J(w)$ change?

$$\begin{aligned}
 w &= 3 + 0.001 \quad \epsilon = 0.002 \\
 J(w) &= w^2 = 9.006001 \\
 &\quad \underbrace{9.012004}_{9.012} \\
 \text{If } w \uparrow 0.001 &\quad \text{If } w \uparrow 0.002 \quad \epsilon \leftarrow \\
 J(w) \uparrow 6 \times 0.001 &\quad J(w) \uparrow 6 \times 0.002 \quad 6 \times \epsilon \quad 6 \times 0.002 = 0.012 \\
 \frac{\partial}{\partial w} J(w) &= 6 \quad \uparrow
 \end{aligned}$$

Informal Definition of Derivative

If $w \uparrow \epsilon$ causes $J(w) \uparrow k \times \epsilon$ then
 $\frac{\partial}{\partial w} J(w) = k \rightarrow 6$

Gradient descent
 repeat {
 $w_j = w_j - \alpha \frac{\partial}{\partial w_j} J(\vec{w}, b)$
 }
 learning rate

If derivative is small, then this update step will make a small update to w_j

If the derivative is large, then this update step will make a large update to w_j

More Derivative Examples

$w = 3$ $J(w) = w^2 = 9$ $w \uparrow 0.001$ $J(w) = J(3.001) = 9.006001$ $\frac{\partial}{\partial w} J(w) = 6$
 $J(w) \uparrow 6 \times 0.001$

$w = 2$ $J(w) = w^2 = 4$ $w \uparrow 0.001$ $J(w) = J(2.001) = 4.004001$ $\frac{\partial}{\partial w} J(w) = 4$
 $J(w) \uparrow 4 \times 0.001$

$w = -3$ $J(w) = w^2 = 9$ $w \uparrow 0.001$ $J(w) = J(-2.999) = 8.994001$ $\frac{\partial}{\partial w} J(w) = -6$
 $J(w) \downarrow 6 \times 0.001$

$J(w) = w^2$

Calculus	w	$\frac{\partial J(w)}{\partial w}$
$\frac{\partial}{\partial w} J(w) = 2w$	3	$2 \times 3 = 6$
	2	$2 \times 2 = 4$
	-3	$2 \times -3 = -6$

Even More Derivative Examples

$w = 2$ $J(w) = w^2 = 4$ $\frac{\partial}{\partial w} J(w) = 2w = 4$ $w \uparrow 0.001$ $J(w) = 4.004001$
 $J(w) \uparrow 4 \times \epsilon$

$J(w) = w^3 = 8$ $\frac{\partial}{\partial w} J(w) = 3w^2 = 12$ $w \uparrow \epsilon$ $J(w) = 8.012006$
 $J(w) \uparrow 12 \times \epsilon$

$J(w) = w = 2$ $\frac{\partial}{\partial w} J(w) = 1$ $w \uparrow \epsilon$ $J(w) = 2.001$
 $J(w) \uparrow 1 \times \epsilon$

$J(w) = \frac{1}{w} = \frac{1}{2} = 0.5$ $\frac{\partial}{\partial w} J(w) = -\frac{1}{w^2} = -\frac{1}{4}$ $w \uparrow \epsilon$ $J(w) = 0.49975$
 $J(w) \downarrow \frac{1}{4} \times \epsilon$

$\frac{\partial}{\partial w} J(w)$	$w \uparrow \epsilon$	$J(w) \uparrow k \times \epsilon$

A note on derivative notation

If $J(w)$ is a function of one variable (w),

$$d \quad \frac{d}{dw} J(w)$$

If $J(w_1, w_2, \dots, w_n)$ is a function of more than one variable,

$$\partial \quad \frac{\partial}{\partial w_i} J(w_1, w_2, \dots, w_n) \quad \frac{\partial J}{\partial w_i} \quad \text{or} \quad \frac{\partial}{\partial w_i} J$$

"partial derivative"

notation used
in these courses

Computational Graph

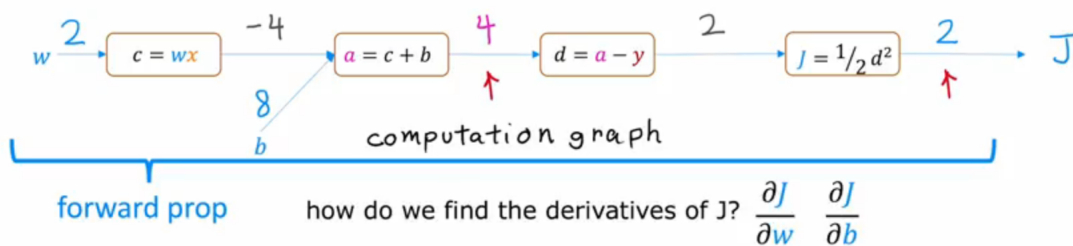
Small Neural Network Example



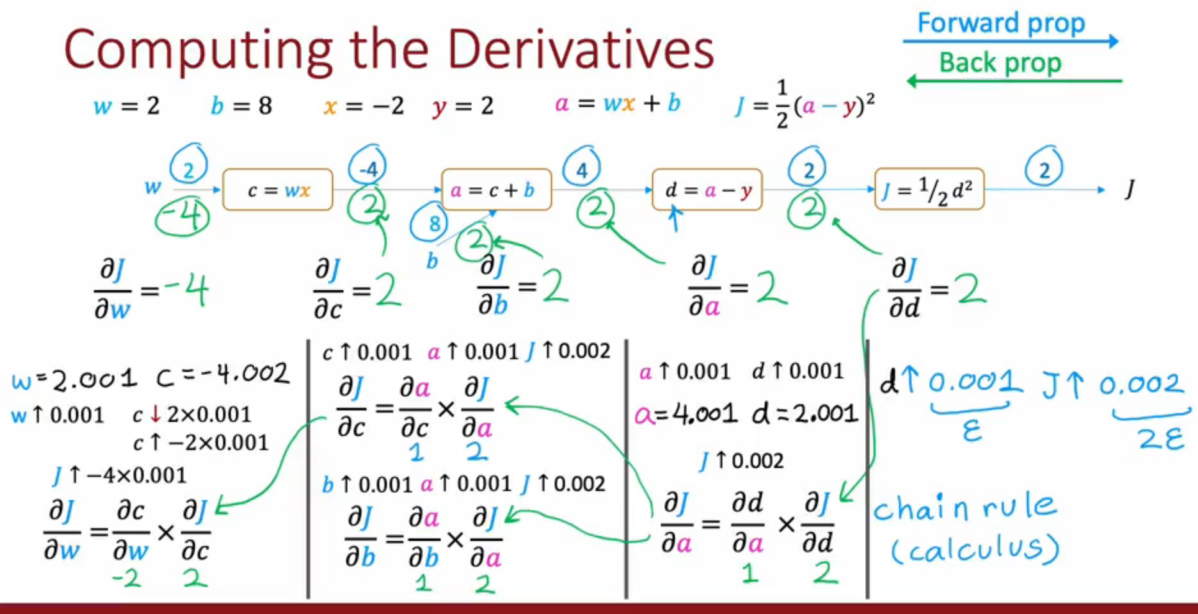
$$w=2 \quad b=8 \quad x=-2 \quad y=2$$

$$a = wx + b \quad \text{linear activation} \quad a = g(z) = z$$

$$J(w, b) = \frac{1}{2} (a - y)^2$$

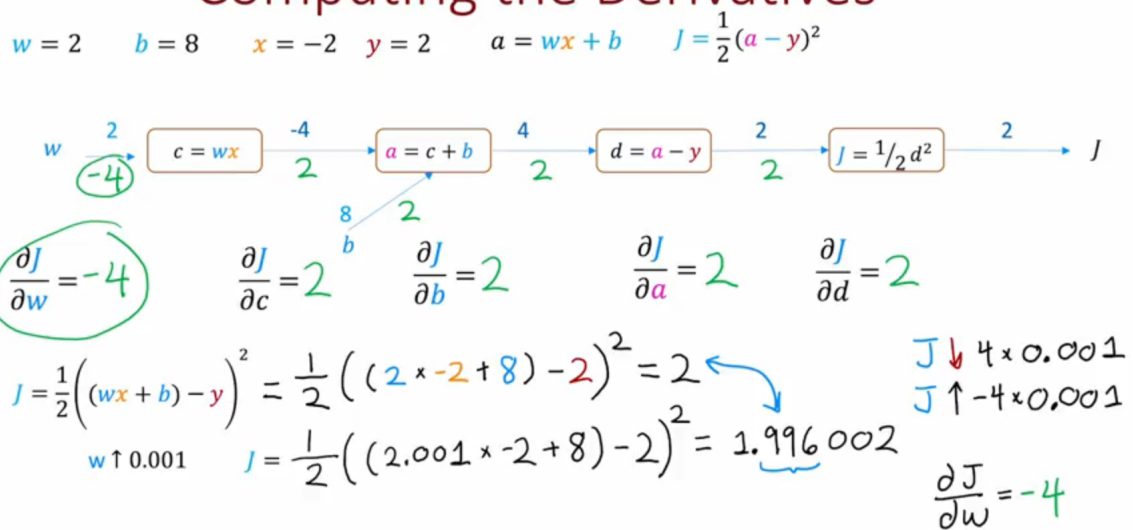


Computing the Derivatives

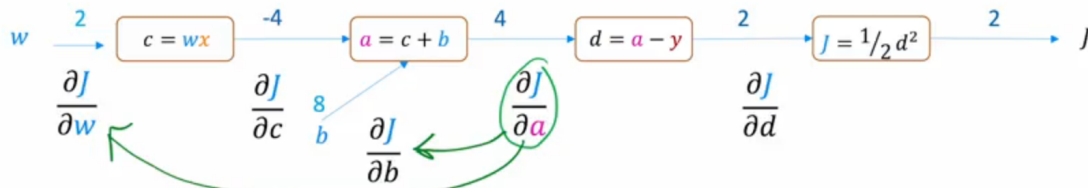


Verify:

Computing the Derivatives



Backprop is an efficient way to compute derivatives

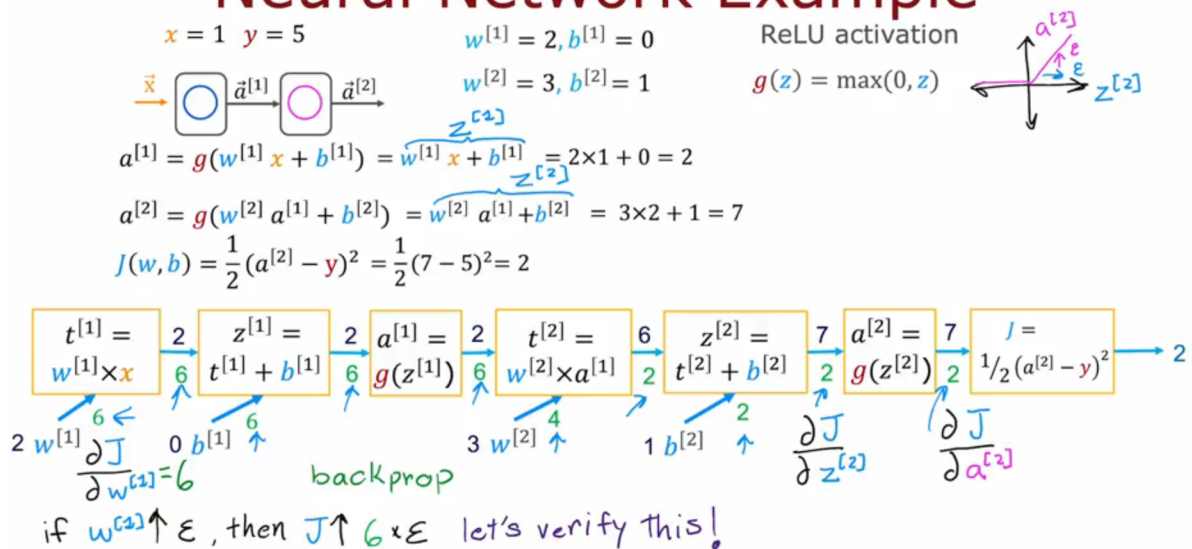


Compute $\frac{\partial J}{\partial a}$ once and use it to compute both $\frac{\partial J}{\partial w}$ and $\frac{\partial J}{\partial b}$.

If N nodes and P parameters, compute derivatives in roughly $N + P$ steps rather than $N \times P$ steps.

N	P	$N + P$	$N \times P$
10,000	100,000	1.1×10^5	10^9

Neural Network Example



Neural Network Example

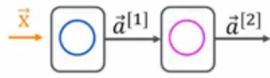
$$x = 1 \quad y = 5$$

$$w^{[1]} = 2, b^{[1]} = 0$$

ReLU activation

$$w^{[2]} = 3, b^{[2]} = 1$$

$$g(z) = \max(0, z)$$



$$a^{[1]} = g(w^{[1]}x + b^{[1]}) = w^{[1]}x + b^{[1]} = 2 \times 1 + 0 = 2$$

$$a^{[2]} = g(w^{[2]}a^{[1]} + b^{[2]}) = w^{[2]}a^{[1]} + b^{[2]} = 3 \times 2 + 1 = 7$$

$$J(w, b) = \frac{1}{2}(a^{[2]} - y)^2 = \frac{1}{2}(7 - 5)^2 = 2$$

$$w^{[2]} \uparrow 0.001 \quad J \uparrow 6 \times 0.001 \quad \frac{\partial J}{\partial w^{[2]}} = 6$$

$$\frac{\partial J}{\partial w^{[2]}} \quad \frac{\partial J}{\partial b^{[2]}}$$

$$\frac{\partial J}{\partial w^{[2]}} \quad \frac{\partial J}{\partial b^{[2]}}$$

N nodes $\square \rightarrow \square \rightarrow \square$

P parameters
 $w_1, b_1, w_2, b_2 \dots$

inefficient way

$N \times P$

efficient way (backprop)

$N + P$